Operator-Specific Forgetting in Matrix-Valued Associative Memory

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Abstract

Selective forgetting is usually evaluated indirectly through retrieval accuracy or overwrite behavior. This work treats forgetting as an operator-specific property of recurrent associative memory, studying a matrix-valued complex memory that stores key–value associations by superposition and retrieves them by matched filtering. Closed-form, fit-free residual laws are derived for two readout channels (complex and real-truncated) under two controlled perturbation locations: key drift and state drift. Under independent random keys, the expected pooled residual depends on the associative load $c = (n-1)/D$ rather than on dimension D alone. Across trained recurrent models, the four predicted laws match pooled residual measurements with maximum absolute deviations of 0.0094 at $D = 64$ and 0.0049 at $D = 128$ under approximately matched load ($c \approx 0.30$). Under coherent key drift, the complex projector produces an algebraically zero residual under the idealized operator (verified numerically below the guard threshold $\varepsilon_{\text{null}} = 10^{-4}$ in all 30 runs at $D = 128$: five seeds $\times$ six drift magnitudes), while other cells leave structured residuals that support linear decoding under the specified probe protocol, even when pooled energy leakage is small. Furthermore, a corrective delta-rule write analyzed under a first-order directional LMS approximation explains its observed divergence when $\beta \|\mathbf{k}\|^2 \geq 2$; normalized-key delta writes restore bounded dynamics and exhibit a small mean selectivity difference (-0.0031) relative to the complex projector, though formal equivalence was inconclusive with five seeds (90% CI [-0.0206, 0.0144]).

1 Introduction

Recurrent associative memories compress a growing sequence of key–value associations into a fixed-size state. This produces a state whose size is independent of sequence length during autoregressive inference, although it remains quadratic in the feature dimension D, aligning with recent linear-attention and recurrent state-space architectures [10, 16, 24, 35, 36]. However, compression introduces interference and makes forgetting difficult to characterize. A system may attenuate an association in energy while leaving a structured residual from which the target remains recoverable by downstream decoders.

This paper addresses a narrower question: when an associative memory is instructed to forget a specific association, what determines whether the erased value remains recoverable? The investigation separates this question into an operator-level residual energy measurement and a functional linear decoding test. It also evaluates two controlled perturbation locations: a phase drift in the erase key at unbinding time, and a write-time phase offset in the target association.

The primary contributions are:

1. A protocol that treats forgetting as an operator-specific benchmark, separating pooled energy leakage from functional decodability.

2. Four closed-form, fit-free residual laws for complex and real-truncated channels under key and state drift.

3. Empirical verification of these laws in trained matrix-valued memories at $D = 64$ and $D = 128$ under approximately matched associative load.

4. An applied LMS reading of the corrective delta write: the classical stability condition $0 < \beta\|\mathbf{k}\|^2 < 2$ [31, 32] explains why an unnormalized corrective baseline diverges in practice, while open-loop superposition writes are unconditionally stable.

The memory construction is grounded in classical associative memory [13, 18, 33], holographic reduced representations, vector-symbolic architectures, complex associative memories, and linear-attention/fast-weight architectures [7–9, 15, 16, 20, 22, 24, 35]. Superposition, conjugate matched filtering, and subtractive unbinding are established ingredients. We formalize operator-specific forgetting, derive closed-form residual laws, and analyze write stability within this memory formulation.

2 Memory and forgetting operators

Let $\mathbf{w}_i \in \mathbb{C}^D$ be fixed codebook keys with unit-modulus entries ($|\mathbf{w}_{i,k}| = 1$, squared Euclidean norm $\|\mathbf{w}_i\|_2^2 = D$) and $\mathbf{v}_i \in \mathbb{R}^D$ be real-valued target vectors. The memory state $\mathbf{M} \in \mathbb{C}^{D\times D}$ stores n associations by superposition:

$$\mathbf{M} = \sum_{i=1}^{n} \mathbf{w}_i \mathbf{v}_i^\top. \tag{1}$$

A matched-filter read with reference key $\mathbf{w}_j$ returns the row vector

$$\hat{\mathbf{v}}_j = \frac{\mathbf{w}_j^H \mathbf{M}}{D} = \mathbf{v}_j^\top + \frac{1}{D}\sum_{k\neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \tag{2}$$

Because the stored targets $\mathbf{v}_i$ are real-valued while the memory state $\mathbf{M}$ is complex-valued, two readout channels are evaluated: the full complex channel $\hat{\mathbf{v}}_j \in \mathbb{C}^D$ and the real-truncated channel $\mathrm{Re}(\hat{\mathbf{v}}_j) \in \mathbb{R}^D$.

We evaluate three model arms:

- **`wave complex`**: complex superposition write, complex reread erase.
- **`wave Re`**: complex superposition write, real-truncated reread erase.
- **`normalized delta`**: normalized-key delta-rule write, projective erase.

All experiments execute erasure through a standardized erase command token (`FORGET_ID`). The primary erase operator is the *reread erase*, which reads an estimate of the target from the current state and subtracts its outer product:

$$\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \hat{\mathbf{v}}_j. \tag{3}$$

For keys with unit-modulus entries and squared norm $\|\mathbf{w}_j\|_2^2 = D$, this operation is algebraically identical to the rank-1 orthogonal projection induced by a $\beta = 1$ delta rule with normalized keys (Section 4). An alternative *representational erase* subtracts the true target value ($\mathbf{M} - \mathbf{w}_j \mathbf{v}_j^\top$), serving as a control condition.

2.1 Controlled drift locations

Two controlled phase interventions are evaluated (Figure 1):

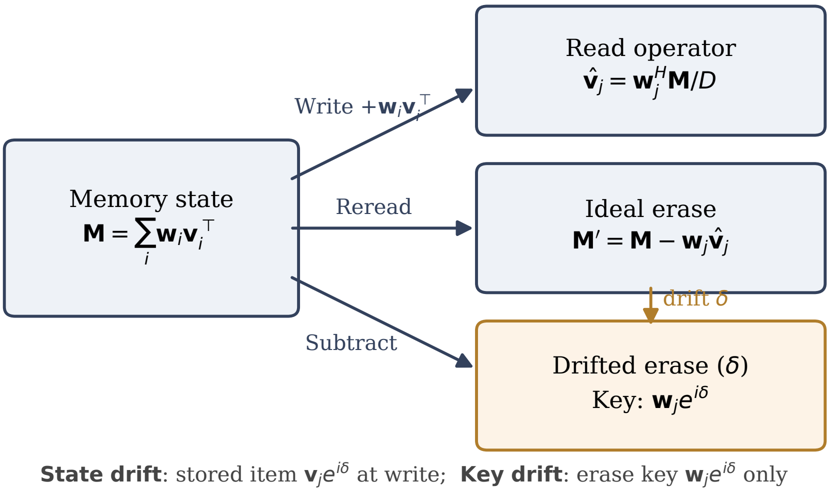

Figure 1: Write, read, and erase operators of the matrix-valued associative memory, illustrating the two controlled drift locations: phase drift of the erase key at unbinding time (key drift) and write-time phase offset on the target association followed by reference-clock mismatch during unbinding (state drift).

1. **Key drift** ($\mathbf{w}_j \to \tilde{\mathbf{w}}_j = \mathbf{w}_j e^{i\delta}$): Memory associations are stored unperturbed. At erase time, local oscillator drift alters the erase key to $\tilde{\mathbf{w}}_j$. Both the re-read estimate $\hat{\mathbf{v}}_j = \tilde{\mathbf{w}}_j^H \mathbf{M}/D$ and the subtractive unbinding use the drifted key:

$$\mathbf{M}' = \mathbf{M} - \tilde{\mathbf{w}}_j \hat{\mathbf{v}}_j. \tag{4}$$

2. **State drift**: We use *state drift* to denote a write-time phase offset attached to the target association, followed by a mismatch between the stored phase coordinate used for re-reading and the reference coordinate used for unbinding. Specifically, the target item is stored as $(e^{i\delta}\mathbf{w}_j)\mathbf{v}_j^\top$. Erase re-reads using the item's stored coordinate $\tilde{\mathbf{w}}_j = e^{i\delta}\mathbf{w}_j$ to obtain value estimate $\hat{\mathbf{v}}_j = \tilde{\mathbf{w}}_j^H \mathbf{M}/D$, but executes unbinding with the unperturbed reference clock key $\mathbf{w}_j$:

$$\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \hat{\mathbf{v}}_j. \tag{5}$$

Query readout is always evaluated with the unperturbed reference key $\mathbf{w}_j$. For the complex channel, the post-erase query residual is $\mathbf{r}_j = \mathbf{w}_j^H \mathbf{M}'/D$. For the real channel, real truncation is applied both during re-read ($\hat{\mathbf{v}}_j \leftarrow \mathrm{Re}(\hat{\mathbf{v}}_j)$) and at query readout: $\mathbf{r}_j = \mathrm{Re}(\mathbf{w}_j^H \mathbf{M}'/D)$. Truncation is applied at both locations because the trained readout head consumes real-truncated representations end-to-end; single-sided variants define different operators and are not evaluated here.

3 Closed-form residual laws

Under the independent random-key model (i.i.d. unit-modulus entries with uniform phases), the dependence on the number of interfering associations enters through $c = (n-1)/D$, following classical distributed capacity scaling [7, 15, 22]. For the pooled residual energy normalized by the pooled target energy $L = \sum_i \|\mathbf{r}_i\|^2 / \sum_i \|\mathbf{v}_i\|^2$, the expected values under random unit-modulus

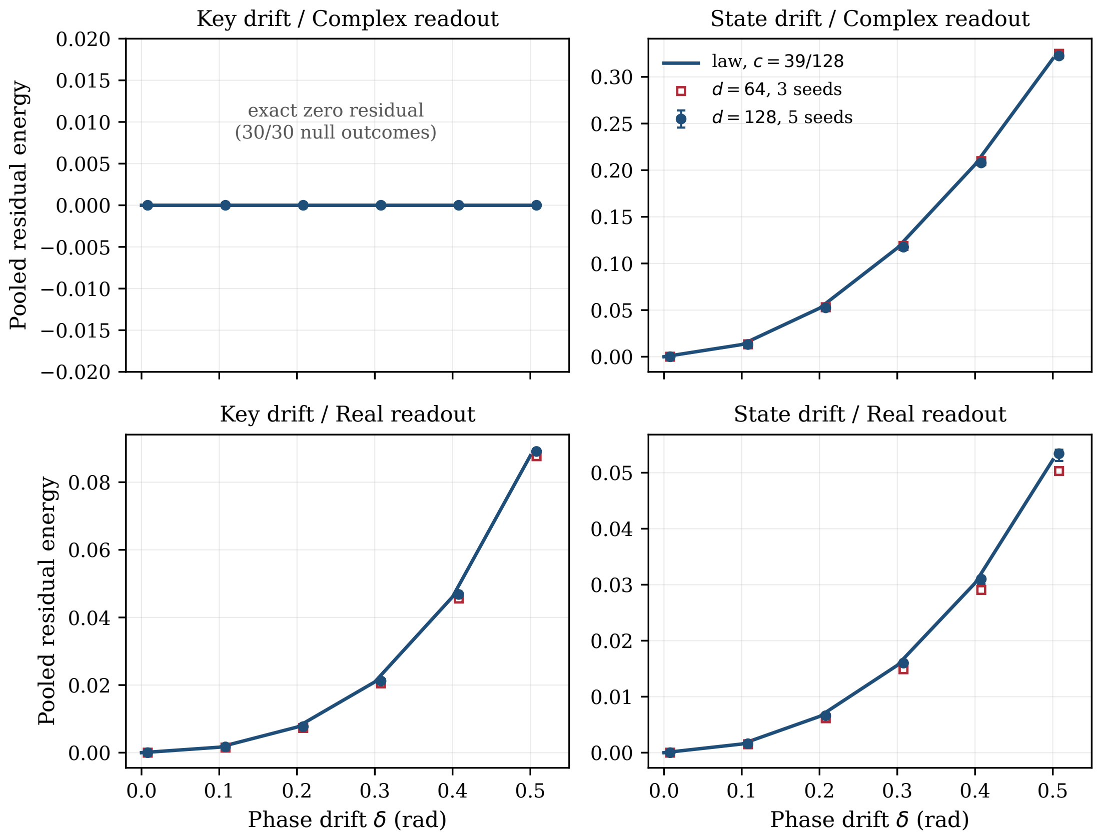

Figure 2: Measured pooled residual energy versus the four closed-form laws across drift values $\delta \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. Closed circles: $D = 128$, five seeds (median with min–max bars). Open squares: $D = 64$, three seeds. The key/complex cell produces an algebraically zero residual under the idealized operator (and numerically null residuals below $\varepsilon_{\text{null}} = 10^{-4}$ across all 30 tested runs: five seeds × six drift magnitudes at $D = 128$).

keys are described by four closed-form, fit-free expressions:

$$L_{\text{key, complex}}(\delta) = 0, \tag{6}$$

$$L_{\text{state, complex}}(\delta) = 2(1 - \cos\delta)\,(1 + c), \tag{7}$$

$$L_{\text{key, Re}}(\delta) = \sin^4\delta + \frac{c}{4}(1 - \cos 2\delta), \tag{8}$$

$$L_{\text{state, Re}}(\delta) = (1 - \cos\delta)^2 + c\,(1 - \cos\delta). \tag{9}$$

Derivation sketch for Eq. (7). Under state drift, the re-read estimate with stored coordinate $\tilde{\mathbf{w}}_j = e^{i\delta}\mathbf{w}_j$ gives $\hat{\mathbf{v}}_j = \mathbf{v}_j^\top + \frac{e^{-i\delta}}{D}\sum_{k\neq j}\mathbf{w}_j^H\mathbf{w}_k\mathbf{v}_k^\top$: the target phase cancels because the estimate is read with the item's own stored coordinate. Subtracting $\mathbf{w}_j\hat{\mathbf{v}}_j$ produces post-erase state $\mathbf{M}' = (e^{i\delta} - 1)\mathbf{w}_j\mathbf{v}_j^\top + \sum_{k\neq j}\mathbf{w}_k\mathbf{v}_k^\top - \mathbf{w}_j\frac{e^{-i\delta}}{D}\sum_{k\neq j}\mathbf{w}_j^H\mathbf{w}_k\mathbf{v}_k^\top$. Reading $\mathbf{M}'$ with reference key $\mathbf{w}_j$ returns target residual $(e^{i\delta} - 1)\mathbf{v}_j^\top$ and crosstalk $(1 - e^{-i\delta})\frac{1}{D}\sum_{k\neq j}(\mathbf{w}_j^H\mathbf{w}_k)\mathbf{v}_k^\top$. Summing squared residual norms over stored items before taking expectations—the level at which the pooled estimator L is defined—and using $\mathbb{E}[|\mathbf{w}_j^H\mathbf{w}_k|^2] = D$ for $k \neq j$ together with $\sum_j\sum_{k\neq j}\|\mathbf{v}_k\|^2 = (n-1)\sum_k\|\mathbf{v}_k\|^2$ yields

$$\frac{\sum_j \mathbb{E}[\|\mathbf{r}_j\|^2]}{\sum_j\|\mathbf{v}_j\|^2} = |e^{i\delta} - 1|^2 + |1 - e^{-i\delta}|^2\frac{n-1}{D} = 2(1 - \cos\delta)(1 + c), \tag{10}$$

with no equal-norm assumption on the target vectors (per-item expectations would additionally require equal target norms and are not invoked).

Key/complex null residual. Under coherent key drift, $\mathbf{M}' = \mathbf{M} - \tilde{\mathbf{w}}_j(\tilde{\mathbf{w}}_j^H \mathbf{M}/D)$ is algebraically an exact orthogonal projection onto the subspace orthogonal to $\tilde{\mathbf{w}}_j$, eliminating the target component identically ($L = 0$ in the algebraic model). In numerical experiments, all 30 runs at $D = 128$ (5 seeds $\times$ 6 drift magnitudes) produced residuals below the pre-specified null threshold $\varepsilon_{\text{null}} = 10^{-4}$. In contrast, the remaining three cells retain structured residuals whose energy alone does not reveal whether a decoder can recover the forgotten value.

Under the independent random-key model, the dependence on the number of interfering associations enters through $c = (n-1)/D$. The expected residual laws are invariant across dimension D at fixed c; empirical trained-model measurements additionally reflect finite-sample and optimization variability (Figure 2).

4 Write stability

The corrective delta write updates a state matrix $\mathbf{S}$ according to

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \beta\,\mathbf{k}_t(\mathbf{v}_t^\top - \mathbf{k}_t^\top \mathbf{S}_{t-1}), \tag{11}$$

where $\mathbf{k}_t$ and $\mathbf{v}_t$ are column vectors; for complex keys and states, transposes are replaced by conjugate transposes. Under a first-order directional approximation assuming quasi-orthogonal keys, the error dynamics along direction $\mathbf{k}_t$ follow the scalar recursion

$$s_t = \left(1 - \beta\|\mathbf{k}_t\|^2\right)s_{t-1} + \beta\,q_t, \tag{12}$$

where q_t represents cross-talk interference. The homogeneous dynamics are stable if and only if $|1 - \beta\|\mathbf{k}_t\|^2| < 1$, which requires $0 < \beta\|\mathbf{k}_t\|^2 < 2$.

With $\beta = 1$ and unnormalized key embeddings initialized such that $\|\mathbf{k}_t\|^2 \approx 8$, the directional feedback multiplier is approximately $1 - 8 = -7$. Because the task draws from a finite key vocabulary that is revisited across updates, this per-direction multiplier acts repeatedly along shared directions; under the quasi-orthogonality approximation these directions evolve independently, and the error components compound so that $\|\mathbf{S}\|$ escalates to 10^{11}–10^{12} within initial epochs, causing the unnormalized corrective baseline to fail. Normalizing the keys ($\|\mathbf{k}_t\|_2^2 = 1$) yields a normalized-key delta update [31, 32]; for $0 < \beta < 2$, this ensures $|1 - \beta| < 1$ and keeps the directional homogeneous dynamics inside the unit disk, maintaining bounded state norm and achieving stable retrieval (Figure 3).

This stability derivation provides an explanatory mechanism for why unnormalized closed-loop delta writes diverge without key normalization. It does not imply that all contemporary delta-rule architectures are unstable, as modern implementations typically incorporate RMSNorm, LayerNorm, or decay gating [35, 36]. In contrast, the open-loop superposition write possesses no corrective feedback loop and does not exhibit the corrective-feedback instability analyzed here.

5 Experiments

A staged evaluation protocol was executed, progressing from synthetic algebra checks to fully trained recurrent models. The scale-matched experiment evaluated $D = 64$ ($n = 20$ pairs, associative load $c = 19/64 \approx 0.2969$, three seeds) and $D = 128$ ($n = 40$ pairs, associative load $c = 39/128 \approx 0.3047$, five seeds) across a drift grid $\delta \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. A discrete vocabulary of 105 tokens was used (48 key tokens, 48 value tokens, and delimiter markers); the token vocabulary is used strictly to construct controlled associative sequences and is not intended as a natural-language modeling benchmark. Each arm and seed evaluated 320 forgetting events. For inferential comparisons, the random seed—not the individual forgetting event—was treated as the independent experimental unit. The unit-modulus key codebook is shared across all seeds (fixed generation seed), so seed-level variability reflects parameter initialization and dataset

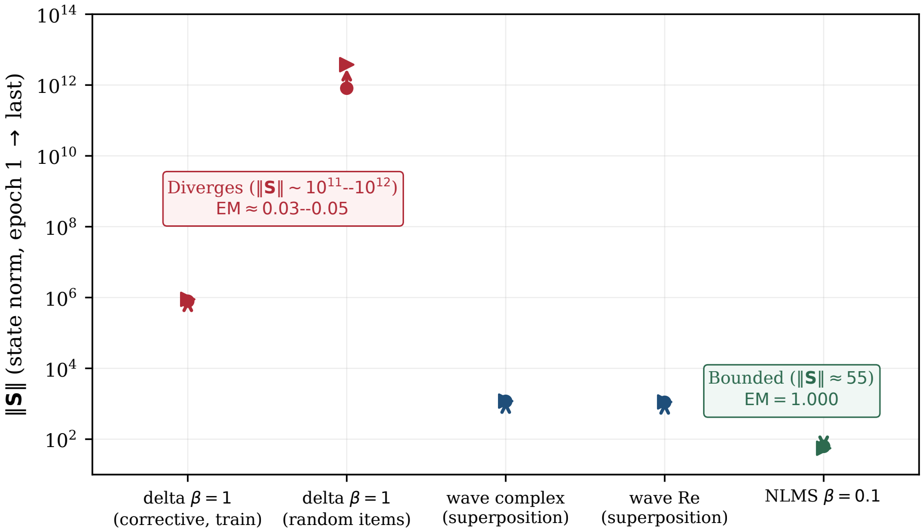

Figure 3: State-norm evolution (log scale, epoch 1 to final epoch). Corrective delta with $\beta = 1$ diverges (EM $\approx$ 0.03–0.05). Superposition writes (wave complex, wave Re) remain bounded. Normalized-key delta with $\beta = 0.1$ maintains bounded state norm and achieves EM = 1.000.

sampling (train/validation pairs and probe events) rather than resampling of the random-key ensemble over which the closed-form expectations are taken; within each run, the probe pools 320 events spanning many query-key realizations.

The primary residual metric was pooled across events:

$$L = \frac{\sum_i \|\mathbf{r}_i\|^2}{\sum_i \|\mathbf{v}_i\|^2}, \tag{13}$$

which avoids the heavy-tail variance of averaging per-example ratios when target vector norms vary. A linear Ridge regression probe ($\alpha = 0.1$, within-model split of 75% train [$n = 240$] and 25% test [$n = 80$]) measured whether the post-erase query residual vector $\mathbf{r}_j$ retained sufficient structure to decode the continuous target vector $\mathbf{v}_j \in \mathbb{R}^D$. At test time, continuous predictions $\hat{\mathbf{v}}$ were mapped to the value vocabulary by nearest cosine similarity to codebook vectors, and exact match was recorded when the recovered token matched the true target. Retrieval selectivity was evaluated across retained and erased pairs:

$$\text{Selectivity} = \text{EM}_{\text{alive}} - \text{EM}_{\text{erased}}. \tag{14}$$

Erased-item exact match (EM) was evaluated only when the residual norm was strictly nonzero; residuals below the guard threshold $\varepsilon_{\text{null}} = 10^{-4}$ were recorded as numerically null residuals. The threshold is set well above the floating-point noise floor of the computation.

At $D = 128$, all 15 seed–arm combinations reached EM = 1.000; at $D = 64$, all 9 seed–arm combinations likewise trained to EM = 1.000. Because the synthetic associative task saturates at this scale, EM serves as a validation sanity check rather than the primary evidence.

6 Results

Table 1 reports the recomputed maximum absolute deviations between measured pooled residuals and the closed-form predictions across cells and dimensions, recomputed directly from the frozen JSON artifacts by `verify_numbers.py`. At $D = 64$, the maximum absolute deviation is 0.0094; at $D = 128$, it is 0.0049. Both values remain below the pre-specified consistency tolerance

Table 1: Maximum absolute deviation |measured − predicted| over seeds and the drift grid per cell, under the pooled estimator (recomputed from frozen artifacts). Key/complex residuals are recorded as numerically null below $\varepsilon_{\text{null}} = 10^{-4}$.

perturbation	readout channel	$D = 64$	$D = 128$
key drift	complex (algebraic zero)	$< 10^{-4}$	$< 10^{-4}$
state drift	complex: $2(1-\cos\delta)(1+c)$	0.0094	0.0049
key drift	real (Re): $\sin^4\delta + c(1-\cos 2\delta)/4$	0.0016	0.0018
state drift	real (Re): $(1-\cos\delta)^2 + c(1-\cos\delta)$	0.0015	0.0018
Pre-specified consistency tolerance		≤ 0.02	≤ 0.02

of 0.02 (Figure 2); ensemble-calibrated standard deviations (Supp. G) place these maxima at approximately one standard deviation of the random-key ensemble.

At approximately matched load ($c \approx 0.30$, differing by 2.7% between dimensions), the three non-trivial cells demonstrate close agreement across dimensions, with a maximum cross-scale difference of 0.0064 (Figure 4). Of this observed maximum, 0.0019 at $\delta = 0.5$ is attributable to the residual load mismatch alone under the state/complex law (reported by `verify_numbers.py`); the remainder reflects finite-sample and optimization variability. Inverting these laws on the measured pooled residuals recovers the generating load $c = (n-1)/D$ at both scales ($c = 39/128 \approx 0.3047$ at $D = 128$ and $c = 19/64 \approx 0.2969$ at $D = 64$); because the laws are linear in c, this inversion is a consistency check rather than independent validation. The key/complex cell is identically zero for every load and therefore carries no information about load dependence; its residuals remain numerically null below $\varepsilon_{\text{null}}$ at both scales.

The key/complex cell produced algebraically zero and numerically null residuals below $\varepsilon_{\text{null}}$ in all 30 tested runs. The remaining cells exhibited non-zero residuals where the residual supported linear decoding under the specified within-model protocol despite low energetic leakage. Specifically, at $D = 128$, the key/Re residual reaches only 0.002 pooled energy at $\delta = 0.1$, yet the ridge decoder recovers the erased value in roughly 5.6% of test events; chance level for nearest-token decoding over the 48-value vocabulary is $1/48 \approx 2.1\%$, so low-δ decoding is only marginally above chance and the dissociation claim rests on the mid-to-high drift range. Decodability remains low until $\delta \approx 0.25$ and increases substantially by $\delta \approx 0.4$, whereas energy increases smoothly across the entire range (Figure 5). This disparity demonstrates why forgetting evaluation requires both operator-level residual metrics and functional decoding probes.

The unnormalized $\beta = 1$ corrective delta baseline failed to train because its write updates diverged (Section 4). Normalized delta writes trained stably and produced a small mean selectivity difference of -0.0031 relative to wave complex, with a two-sided 90% confidence interval (TOST at $\alpha = 0.05$; per-seed selectivity differences paired across arms, Student-t interval with four degrees of freedom) of $[-0.0206, 0.0144]$ (Figure 6). The two arms are compared under their respective deployed training recipes, which differ in budget (Section 5; Supp. F), so the comparison concerns operator-plus-recipe composites; isolating the write operators alone would require budget-matched retraining. Because the lower confidence bound crosses the equivalence margin $-\varepsilon = -0.02$ by 0.0006 under $n = 5$ seeds, the result is reported as inconclusive with respect to formal equivalence within margin $\varepsilon = 0.02$.

7 Relation to prior work

The memory construction sits within a long associative-memory lineage: correlation-matrix memories [18, 33], the Hopfield network and its signal-to-noise analyses of crosstalk [13], dense associative extensions [19], and the modern Hopfield–attention bridge [23]. Within this tradition,

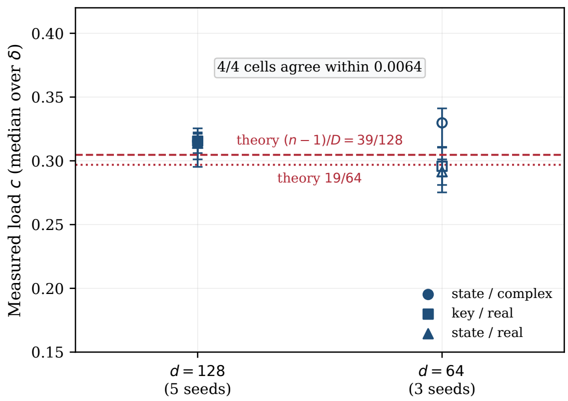

Figure 4: Measured load c (median over drift values $\delta > 0$, with min–max bars) inverted from the three non-zero laws versus expected load $(n-1)/D$. Cross-dimensional agreement: maximum cell difference 0.0064 for the three non-trivial cells.

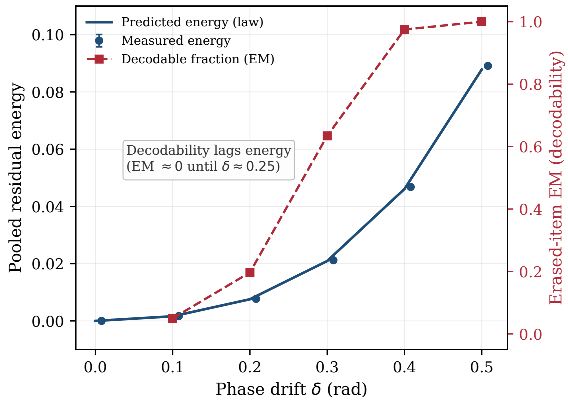

Figure 5: Key/Re cell at $D = 128$: pooled residual energy versus erased-item EM (ridge decodability). Decodability lags energetic leakage: remaining near zero until $\delta \approx 0.25$ and increasing substantially by $\delta \approx 0.4$, whereas energy increases smoothly across the entire range.

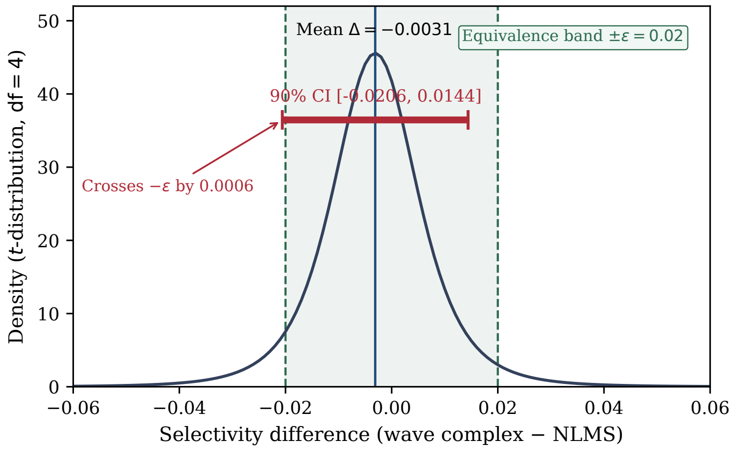

Figure 6: Two one-sided t-test (TOST at $\alpha = 0.05$, 90% CI) on retrieval selectivity between wave complex and normalized delta ($\varepsilon = 0.02$, $n = 5$ seeds). Mean difference: -0.0031; 90% CI: $[-0.0206, 0.0144]$. The confidence interval crosses $-\varepsilon$ by 0.0006.

the residual laws of Section 3 are second-moment statements about crosstalk, in the same species as classical SNR computations; what is added here is an operator- and channel-resolved characterization of what erasure leaves behind. The superposition–matched-filter–unbinding formalism is holographic reduced representation and VSA machinery [9, 15, 22], with capacity characterized by Frady et al. [7] and factorization analyzed by Frady et al. [8]; complex-valued phasor memories trace to Noest [20] and reappear in recent phase-associative sequence models [30], which share the complex outer-product state but target language modeling at scale rather than erase-residual characterization.

On the fast-weight side, fast weights were introduced by Schmidhuber [25] and revisited as attention by Ba et al. [3]; linear attention was established by Katharopoulos et al. [16], with the delta rule cast as a fast-weight programmer by Schlag et al. [24]. Delta-rule linear attention and its parallel forms [35], multiplicative decay architectures—RWKV [21], RetNet [29], Mamba and structured state-space duality [6, 10], gated linear attention [34]—constitute the modern landscape in which forgetting is implemented primarily through multiplicative decay gating rather than subtractive erasure. Two test-time-memory lines are adjacent but mechanistically distinct: TTT layers update their memory state by gradient steps at inference time [28], and Titans adds a surprise-gated neural long-term memory alongside attention [4]. Two fine-grained delta-rule lines are closest to this work: Kimi Delta Attention extends Gated DeltaNet [36] with finer-grained decay gating [17], and Comba analyzes bilinear delta-rule recurrences through closed-loop control theory [14], a control-theoretic reading complementary to the LMS analysis of Section 4. Gated DeltaNet-2 [12] further decouples erasing from writing through channel-wise erase and write gates; the present laws characterize subtractive projection-style erasure, and deriving their analogues under multiplicative, channel-wise erasure is an open direction. As a distant analogue in deployed systems, rotary position embeddings attach position-dependent phases to keys whose write-time and read-time values differ [27]; unlike the controlled drift studied here, those rotations are deterministic and compensated at readout. Recall-centric evaluations in linear-attention systems [1, 2] complement the operator-level protocol studied here. Classical LMS/NLMS adaptive filtering theory [31, 32] provides the foundation for the directional stability analysis in Section 4.

The framework is also related to machine unlearning and privacy research [5, 11, 26], but addresses a different problem: machine-unlearning methods typically evaluate parameter-space indistinguishability from a retrained model, whereas the present study measures residual state structure and separates energetic leakage from functional recoverability in recurrent memory states.

The scope of novelty is specifically delimited: establishing a forgetting-centric evaluation protocol, formalizing the divergence between energetic and functional leakage, deriving the four-way drift/channel residual laws, and empirically validating these closed-form laws in trained recurrent memories under approximately matched associative load.

8 Limitations

The present study utilizes synthetic key–value associative tasks, fixed codebooks, real-valued targets, an explicit matrix state requiring $O(D^2)$ parameters per layer, and controlled drift interventions. The closed-form laws assume independent unit-modulus random keys and single-item phase perturbations; learned codebooks, correlated multi-item drift, natural-language sequence modeling, and hardware throughput benchmarking represent avenues for future research. Statistical comparisons between arms are limited by the available seed count ($n = 5$ at $D = 128$, $n = 3$ at $D = 64$); no prospective power analysis was conducted, although a post-hoc calculation from the observed dispersion (per-seed standard deviation ≈ 0.018) indicates that $n \approx 8$ paired seeds would provide approximately 90% power to establish equivalence at the margin $\varepsilon = 0.02$. The cross-scale comparison holds the associative load approximately matched (within 2.7%; exact matching would require $n = 39$ pairs at $D = 128$), and a systematic sweep of c at fixed D has not been performed. Erasure through multiplicative, channel-wise decay or erase gates—as deployed in recent gated delta architectures [12, 36]—is not modeled here; the present laws characterize subtractive projection-style erasure. These findings do not claim general superiority over Transformer self-attention or all linear-attention architectures.

9 Conclusion

Selective forgetting in recurrent associative memory cannot be characterized adequately by retrieval accuracy alone: energy leakage and functional recoverability diverge by more than an order of magnitude in some cells. Within fixed-codebook matrix-valued memories, what erasure leaves behind is governed by four closed-form laws with no free parameters, verified on trained models at two scales under approximately matched load; exact erasure is a property of the projector acting with a fresh key rather than of the operator family. Two boundaries define the present scope: erasure here is subtractive projection, whereas deployed gated architectures forget primarily through multiplicative decay; and the equivalence between superposition and normalized-delta operators remains statistically open at five seeds. Closing both—multiplicative-erasure residual laws and adequately powered equivalence testing—is the natural continuation of this work.

Reproducibility

Code, configurations, checkpoints, raw JSON outputs, execution logs, and verification scripts are released in the open repository `https://github.com/01javierescobar/operator-specific-forgetting`, archived with DOI 10.5281/zenodo.22051928. The frozen reference commit is `cec4ffe`, tagged `lab-radioactivo-v1`. All quantitative values in this manuscript are recomputed directly from the frozen JSON artifacts by `paper/verify_numbers.py`; all figures are regenerated by `paper/make_figures.py`. The primary trained-model gate (O05) was executed on two NVIDIA T4 GPUs (Kaggle kernel `javierezcobar/onda-o05`, 5.44 h, 15 runs).

Use of AI tools

AI-assisted tools (large language models) were used throughout the research process for code development, experimental design iteration, debugging, and adversarial verification of intermediate claims across the O01–O05 protocol. All mathematical derivations, experimental protocols, quantitative results, and scientific conclusions were validated by the author. The AI tools are not listed as authors and bear no responsibility for the content of this manuscript.

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